



Dry lab notebook

DHCR7 enzyme screening

I. Preparation

[1/07/2023 - 20/07/2023]

Motivation

The final stage of the campesterol biosynthesis pathway requires introducing a gene coding delta-7-sterol reductase since it's not present in *Yarrowia lipolytica*. There are multiple organisms that have the DHCR7 gene so they could potentially be used in our project. The question remains, which of these organisms shows the most potential for our purpose? In order to investigate that, we decided to perform screening of the DHCR7 enzymes.

By contrast to a traditional screening, instead of looking for the best molecule, we were looking for the best receptor. We composed a list of 10 organisms that have the delta-7-sterol reductase coding gene. More on that can be found in the next section of the notebook. In an effort to determine which of these enzymes are the most promising, we performed molecular docking and molecular dynamics simulations of the selected docked conformations.

List of enzymes for screening

In order to compose the list of enzymes for screening we did a literature search. Previous literature^{1,2} lists DHCR7 enzymes from several organisms. According to the studies in these articles, DHCR7 enzyme from *Danio rerio* showed highest yield of campesterol. In our screening, we included all enzymes studied in these articles and homologs of DHCR7 from *Danio rerio* (the ones previously not included).

Our screening includes DHCR7 enzymes from 10 organisms:

- *Rattus norvegicus*
- *Mus musculus*
- *Homo sapiens*
- *Danio rerio*
- *Xenopus laevis*
- *Xenopus tropicalis*
- *Gallus gallus*
- *Waddlia chondrophila*
- *Arabidopsis thaliana*
- *Bos taurus*

From literature: *Rattus norvegicus*, *Xenopus laevis*, *Homo sapiens*, *Gallus gallus*, *Danio rerio*, *Arabidopsis thaliana*, *Waddlia chondrophila*



Additional *Danio rerio* homologs: *Mus musculus*, *Xenopus tropicalis*, *Bos taurus*

Note on *Oryza saliva*: Os02g0465400 protein was studied in one of the listed articles but it showed substantial difference in the structure compared to other enzymes which poses problems for example in locating the binding site for NADPH and ergosta-5.7-dienol. Additionally, even though there is a CSM of this enzyme on the Protein Data Bank (PDB), there is no reviewed record of this protein on UniProt.

Homologs from organisms *P. troglodytes*, *C. lupus*, *M. oryzae* were not included in our study due to removal of the record or unreviewed status.

Due to DHCR7 enzyme being a transmembrane protein and difficulties with structure determination of these proteins, there is no experimental structure of any DHCR7 enzyme available. 4QUV is a delta(14)-sterol reductase which belongs to the same family as DHCR7. It also shows high structural similarity. This structure was used in the docking as a “reference” structure that helped us locate the binding site of NADPH in DHCR7.

For enzymes from all organisms besides *Gallus gallus* and *Waddlia chondrophila*, the CSMs were obtained from the PDB database. Structures of DHCR7 enzyme from *Gallus gallus* and *Waddlia chondrophila* were not deposited to the PDB database, hence we predicted these structures using AlphaFold³.

II. Structure prediction

[28/07/2023 (AlphaFold Colab), 2/08/2023 (ColabFold)]

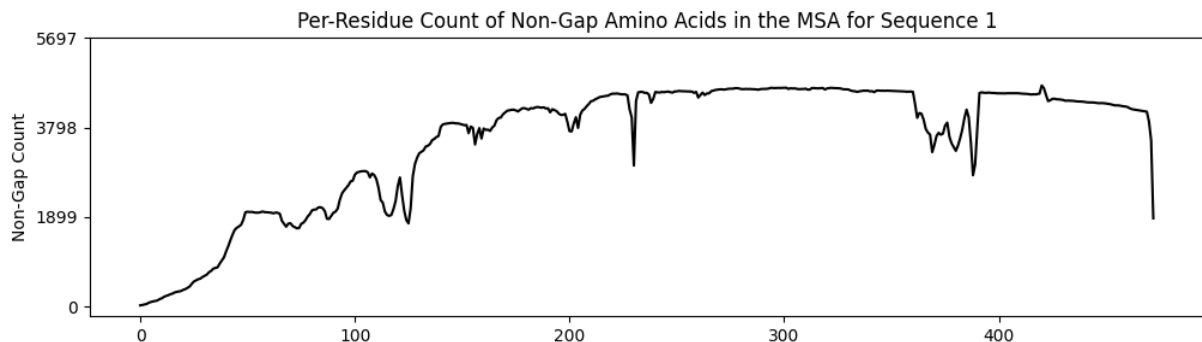
In order to obtain the structure of the DHCR7 enzyme from *Gallus gallus* and *Waddlia chondrophila* we used AlphaFold as a prediction tool. In particular, we used AlphaFold Colab³ and ColabFold⁴ tools on Google Colab and then we compared the results from both tools. All parameters were set with default values.

We decided not to use the full version local AlphaFold because we believe that it won't improve our results that substantially since AlphaFold Colab and ColabFold already provided very good results and predictions with high confidence.

Gallus gallus:

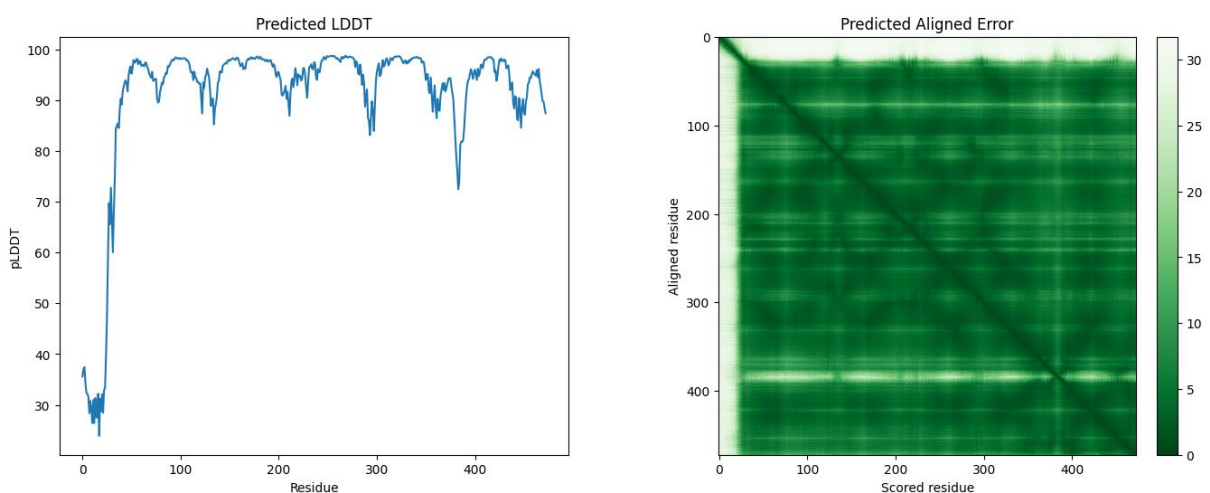
AlphaFold Colab

The first figure shows the residue coverage in the multiple sequence alignment (MSA). Overall, it seems high, there are some drops of the residue count but they usually correlate with pLDDT drops (like the one around residue 380).



Predicted LDDT indicates the confidence of the prediction. pLDDT for DHCR7 in *Gallus gallus* is in general very high, pLDDT higher than 90 points to high accuracy and areas with pLDDT between 70 and 90 indicate to be modelled well. Most of the residues have pLDDT higher than 80 which shows at least well prediction. N-terminus shows lower confidence which is expected since it can be more mobile. In this case the N-terminus is an unstructured region (which is also the case in most of the predicted DHCR7 structures from different organisms).

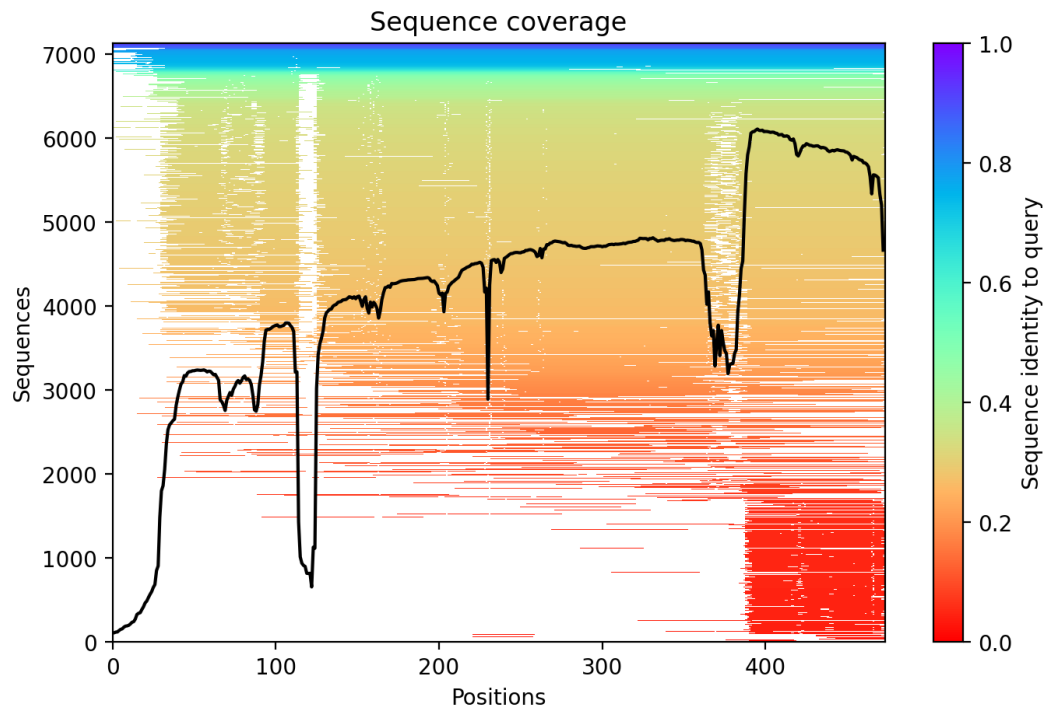
Predicted Aligned Error indicated uncertainty of relative positions and/or orientations of domains that the residues are in. The lower the PAE, the better the prediction in specific regions. Regions with higher PAE (20-30) correspond to the regions with lower pLDDT.



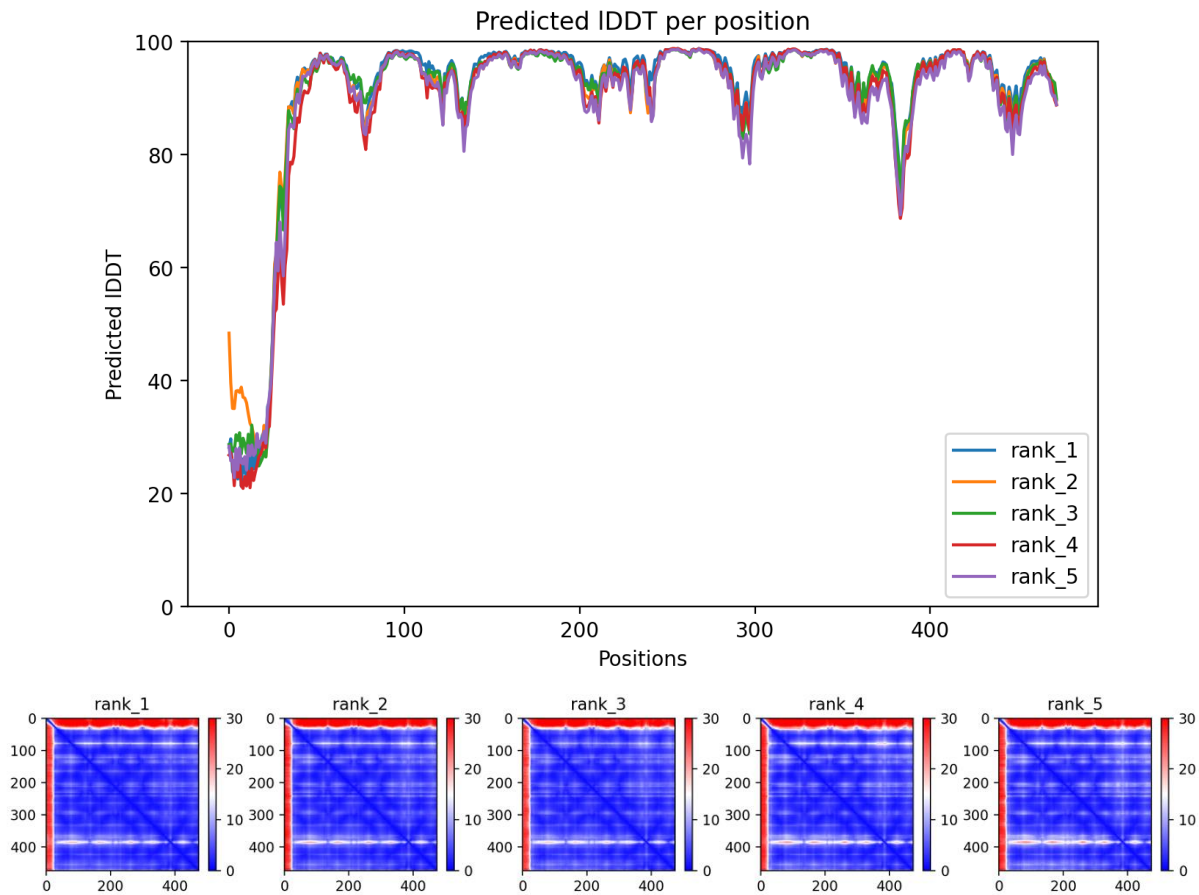
ColabFold

ColabFold returned 5 predicted structures with similar pLDDT and PAE values. Sequence coverage in MSA is quite high, especially for the last 90-100 residues. There are some drops in the coverage around

residues 120 and 380 (the same as in AlphaFold Colab). AlphaFold Colab seems to find more sequences for the region around residue 120.



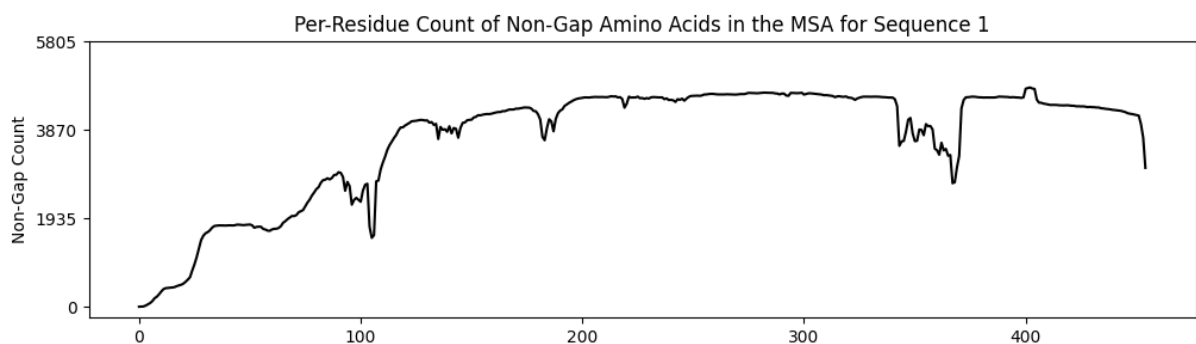
Predicted LDDT values for all 5 models are very similar, model 2 seems to have the most different pLDDT in the N-terminus region. As expected, the N-terminus of the model shows lower confidence. Overall confidence level is well (above 80) with a slight drop in the region around residue 380 (correlated to lower coverage and seen as well in the AlphaFold Colab results). Highest PAE (around 20-30) corresponds to the N-terminus and the relatively high PAE (around 15) to the region around residue 380.



Waddlia chondrophila:

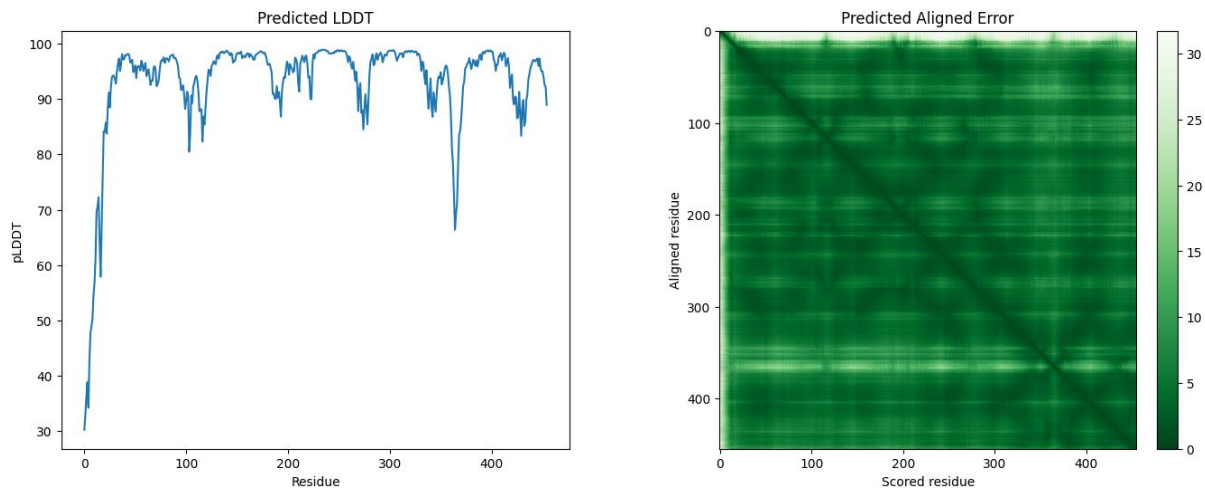
AlphaFold Colab

The sequence coverage in the MSA is relatively high throughout the whole sequence (not taking the N-terminus into account). There is lower coverage in region around residues 350-370 – it could correspond to the same region as in *Gallus gallus* when aligned.



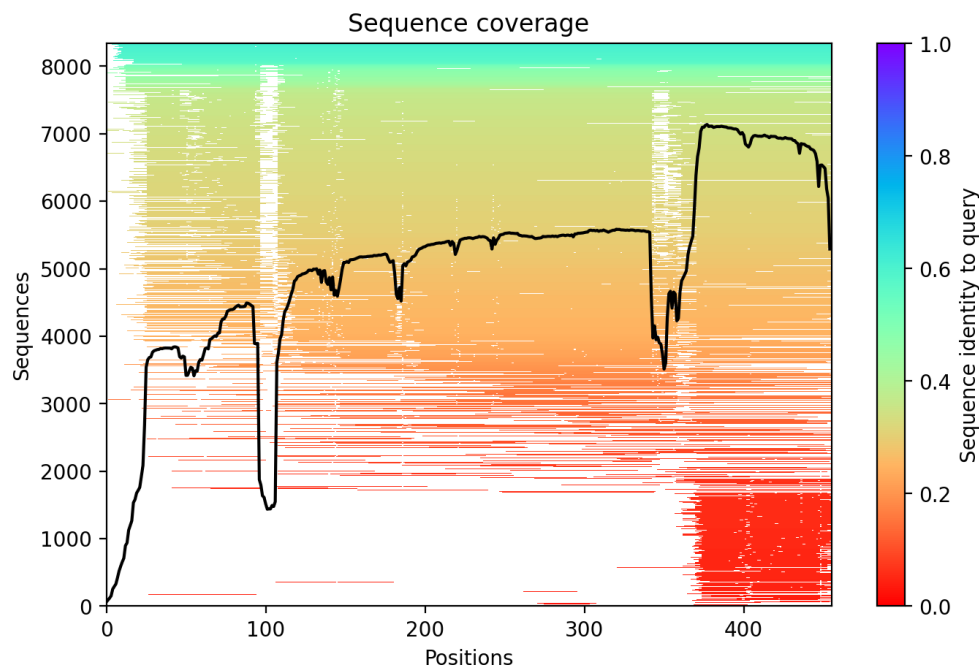
Predicted LDDT is a bit lower than for *Gallus gallus* but the structure is expected to be modelled well. N-terminus displays lower pLDDT as seen previously. Substantial drop of pLDDT is visible around residues 360-370 (probably related to lower coverage in that region). This region's pLDDT drops below

70, therefore it should be treated with caution. Higher PAE values also correspond to either the N-terminus or region around residues 350-370.



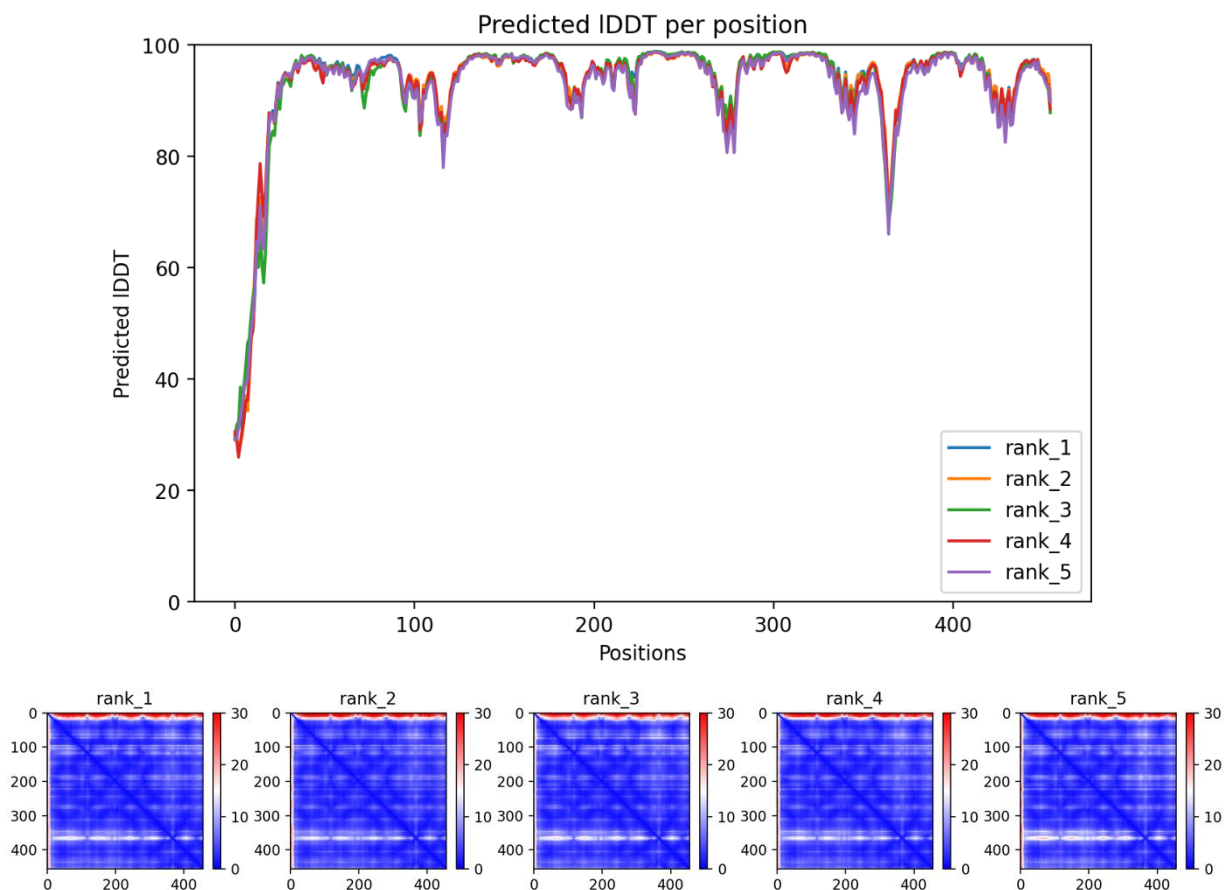
ColabFold

ColabFold returned 5 predicted structures with similar pLDDT and PAE values. Sequence coverage in MSA is quite high and similarly as in *Gallus gallus*, especially high for the last 90-100 residues and the coverage around residues 100 and 350-370 drops. These regions could structurally correspond to the same regions with lower coverage in *Gallus gallus*.



The pLDDT values for all 5 models are very similar. The confidence of prediction for most of the residues is very high (> 80) besides the N-terminus (as expected) and the region around residues 350-

370 which should be interpreted with caution. Highest PAE (around 20-30) corresponds to the N-terminus and the relatively high PAE (around 15) to the region with lower confidence (residues 350-370).



The results from both tools are very similar but based on slightly better pLDDT values we decided to use structures obtained from AlphaFold Colab.

III. Molecular docking

[1/08/2023 - 15/09/2023]

For docking we followed a protocol provided by prof. Jeremy Harvey. We used AutoDock Vina⁵ in UCSF Chimera⁶.

As validation of the protocol, we performed docking of the experimental structure of delta(14)-sterol reductase (4QUV) with its native ligand (NDP). The ligand docked in the same place as in the crystal structure. During docking NDP with 4QUV we encountered a problem – one hydrogen atom in the ligand file was called “HO” for whatever reason (after DockPrep), and Vina assumed it was a Holmium atom, which it was not. After consultation with prof. Harvey, we figured out the problem and changed the name of the hydrogen back to “H” instead of “HO”.

Ligands for further docking (NADPH and ergosta-5.7-dienol) were obtained from PubChem. We perform docking of both NADPH and ergosta-5.7-dienol to each enzyme 10 times in order to get more results.

Viewing docking results in Chimera:

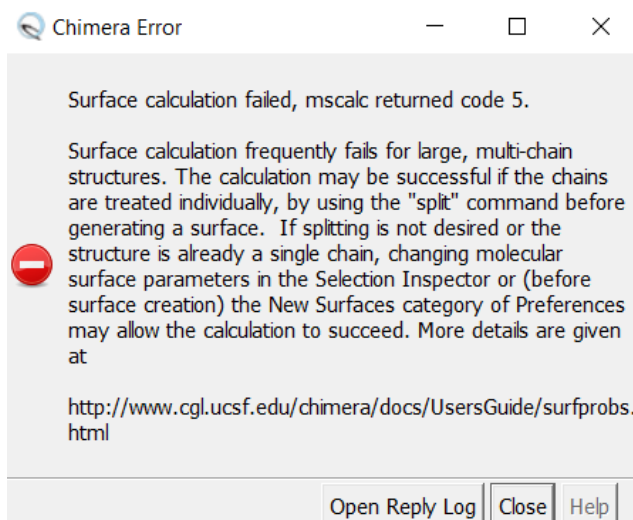
1. Open your enzyme (preferably the one in .pdb format) [File -> Open]
2. Open the docking results with ViewDock [Tools -> Surface/Binding analysis -> ViewDock and load the docking result file in .pdbqt file format]
3. [Optional] You can ungroup the docking conformations [Tools -> General controls -> Model Panel, click on the docking result row, and click group/ungroup]

Since docking scores given by Vina are not always a good monitor of good docking, we also performed double checking to support indication given by the binding energies. Prof. Matheus Froeyen suggested that we look at hydrogen bonds, electrostatic interactions and hydrophobic contacts.

Double checks in Chimera:

- Hydrogen bonds
 - o Compute hydrogen bonds [in ViewDock window go HBonds -> Add Count to Entire Receptor or Tools -> Surface/Binding analysis -> FindHBond]
 - i. Check option inter-model
 - ii. Change the H-bond color to yellow [click on the color box and type *yellow* in the color name box]
 - iii. Set line width to 3
 - iv. Check box "label H-bond with distance"
 - v. Check box "write information to file"
 - vi. Leave rest with default options (as it is, don't change anything else)
 - vii. Click OK or Apply
 - o Show all hydrogen bonds
 - i. Select bonded atoms [Tools -> General controls -> PseudoBond Panel -> select bonded]
 - ii. Show hydrogen bonds and side chains of involved residues [Actions -> Atoms/Bonds -> side chain/base -> show]
 - o Inspect the hydrogen bonds in all docked conformations, check the distance of the length or if there is something suspicious (like no bonds at all or crazy long bond)
- Electrostatic interactions
- Hydrophobic contacts

Note on electrostatic interactions and hydrophobic contacts: When trying to show the surface of the enzyme, Chimera returns an error (screenshot below). The workarounds suggested in the user guide did not help and using ChimeraX (also as suggested in the user guide) generates other errors.



After discussing these problems with prof. Froeyen, he suggested using LigPlot+⁷ and PoseView⁸ to inspect the interactions between the protein and docked ligands.

Double checks with PoseView and LigPlot+:

- Select conformations to investigate and save separate PDB files
 - o Open all docking results (see instructions above for that), preferably open all 10 docking results at once (many windows will pop up, don't worry about that)
 - o Pick 5 conformations from all docking results for each organism (look at best scores and pick a conformation that is lodged similarly as the ligand in our reference structure – NDP in 4QUV)
 - o Make a folder for the selected conformations (e.g., *selected_docking*)
 - o For each conformation save a PDB file [in Chimera window (not the ViewDock window) go to File -> Save PDB...]
 - i. Give a file name (include the number of the docking result and the number of the conformation, e.g., *docking1-1.pdb* or *docking3-5.pdb*)
 - ii. **In "Save models" select the receptor and the picked conformation**
 - iii. In "Save relative to model" pick the receptor
 - iv. In "Save multiple models in": a single file
 - v. Click Save
- PoseView
 - o Prepare the files:
 - i. Make a dedicated folder for PoseView in the *selected_docking* folder
 - ii. Copy the PDB file to this folder
 - iii. Open the PDB file in a text editor

- iv. Find the “ENDMDL” line between the protein and the ligand and delete it
 - v. Make sure to save the file
- o Open the [PoseView](#) page
- o Upload the PDB file of one of the selected docking conformations [Upload protein -> Select file]
- o Click GO
- o Select PoseView from the panel on the right and open it
- o Write the name of the ligand [Settings -> Ligands]; make sure the name of the ligand is exactly the same as the name from the list and make sure there is no space before and/or after the name
- o Click Calculate and wait for results
- o Check the interactions found by PoseView (interpreting the results will be mentioned below)
- o Save the svg file in the *selected_docking* folder (even in the dedicated folder for PoseView), include the name of the PDB file given earlier (e.g. *docking1-1.pdb* or *docking3-5.pdb*)
- LigPlot+
 - o Prepare the files:
 - i. Make a dedicated folder for LigPlot in the *selected_docking* folder
 - ii. Copy the PDB file to this folder
 - iii. Open the PDB file in a text editor
 - iv. Find all lines that start with “MODEL” and “ENDMDL” and delete them (don’t delete the “END” line at the end of the file)
 - v. Make sure to save the file
 - o Open LigPlot+
 - o Open the PDB file of one of the selected docking conformations [File -> Open -> PDB file -> Browse]
 - o Select ligand to plot (click on the ligand in the table)
 - o Click Run
 - o Save the plot [File -> Save] in the *selected_docking* folder (even in the dedicated folder for LigPlot), include the name of the PDB file given earlier (e.g. *docking1-1.pdb* or *docking3-5.pdb*)

Interpreting results from PoseView & LigPlot+:

1. No interactions at all = BAD (we want to have some interactions between our ligand and receptor in order for them to bind specifically)
2. Some hydrogen bonds = GREAT (hydrogen bonds mean that the molecule is anchored in a certain position or orientation and that it will bind specifically)
3. [If there are any hydrogen bonds] Open the UniProt record of the enzyme from the specific organism and look at the “Binding site” section. If the hydrogen bond is formed with one of the residues listed in the UniProt binding site section then it’s AMAZING (it means that our molecule anchored itself specifically to one of the confirmed binding site residues)



4. Hydrophobic contacts = OKAY (it's not as good as presence of hydrogen bonds but hydrophobic contacts also stabilize the molecule and contribute to the binding strength)

IV. Molecular dynamics simulations

[19/09/2023 - 9/09/2023]

Molecular dynamics simulations were performed using AMBER software. Structures for simulations were selected from previously performed molecular docking with double checks (looking at interactions).

1) Preparation of the small molecule

First, the ligand is downloaded from the PubChem database in the SDF format and converted to the PDB format.

Command:

```
./Run.sdf2pdb [ligand_filename]
```

The converted pdb file does not have unique atom names (3rd column) so they need to be changed because AMBER might complain about that later. New names should contain the atom type (O, H, N) and the atom number (01, 16, 72, etc.). Example: the 1st atom is an oxygen, it's new name should be O01; 12th atom is a hydrogen, it's new name should be H12. In addition, the residue name should be changed to LIG (4th column). This is done with the use of script *Run.restorepdb*.

Command:

```
./Run.restorepdb [ligand_filename.pdb] [ligand_filename.out.pdb]
```

Next, the ligand is prepped using programs *antechamber* and *parmchk2* both with net charge 0. The programs generate AMBER force field parameters. Here we used the general AMBER force field 2 (GAFF2).

Commands:

```
./Run.antechamber2 [ligand filename (without .pdb)] 0
```

```
./Run.parmchk2 [ligand filename (without .pdb)] 0
```



2) Preparation of the docked complex:

From the PDB file of the chosen complex lines containing "MODEL" and "ENDMDL" are removed (no blank lines in the file). Additionally, the lines mentioning the secondary structure (helix, sheet) are removed.

The numbering of the atoms in the ligand can differ between the docked ligand and the prepped ligand. Both ligand structures were investigated and the atom names (numbering in particular) in the docked conformation were adjusted to the prepped ligand atom names. Everything was then checked (atom by atom) in Chimera.

The complex file with these modifications is called *[complex filename].out.pdb*.

Additionally, because our structures were predicted by AlphaFold, most of them have a disordered C-terminal. In order to save some computational time, we cut the "tail" (around 20aa) since it's not functional anyways.

LEaP:

The LEaP program generates an AMBER topology file and coordinate file for a structure given a particular force field. We used LEaP with the use of *Run.leap.sol* script. In order for script to work, lines 18 and 19 have to be filled with correct ligand files and line 21 has to contain the correct complex pdb file (in our case *[complex_filename].out.pdb*).

Command:

```
./Run.leap.sol
```

Note: Program can complain about some protonated histidines -> open PDB in Chimera, select hydrogens (HD1) in histidines and delete them. Save the file to PDB and delete the lines mentioning the secondary structure.

3) MD simulation:

Minimization:

In order to relax the structure, first we perform energy minimization. The minimization is run with the use of *Run.min.cuda* with default parameters on the GPU. The script requires changing the residue numbers in the mask.

Command:

```
./Run.min.cuda
```



Heating:

In the heating stage, the system is slowly heated up to the desired temperature. Our system is heated up to 300K with 2fs timestep. Same as in the minimization script, the number of residues need to be updated in the *Run.heat.cuda* script.

Command:

```
./Run.heat.cuda
```

Density:

To adjust the density of the system, we run the *Run.density.cuda* script with updated number of residues in the mask.

Command:

```
./Run.density.cuda
```

Equilibration:

The final step before production is equilibration. Our equilibration was performed with 2500000 steps and 2fs timestep which calculates to a 5000ps (5ns) equilibration run. Since there are no restraints here, there is no mask, hence there is no need to adjust the residue number in the script.

After running the equilibrium, you can check the system by converting the *equil.rst* file to a PDB file and opening it in a visualization software (Chimera, PyMOL, etc.) and see if nothing unfolded or did anything weird happen.

Commands:

```
./Run.equil.cuda
```

```
./Run.makepdb equil
```

After running the script *Run.makepdb* the file *equil.pdb* is created, it can be open and analyzed in any visualization tool (just like any other PDB file).

(Conversion of a *.rst* file into a *.pdb* file can be done at any point during the analysis, not just after equilibration)

Production:

Production is the longest part of the MD simulation. It's best to split production phase into few stages with fewer steps instead of running the whole production at once. This way it's easier to monitor the system and if the job crashes at some point, there is no need of rerunning everything.

Each stage of our production phase includes 2 million steps (with 2fs timestep) which sums up to 4ns. In total, we performed 5 production stages which translates to 20-ns MD simulation.



It is vital to monitor the RMSD after each production stage to ensure the stability of the system. Constant increase in the RMSD throughout the simulation suggests that the simulation time should be increased. Production can be stopped after reaching a stable state which is reflected by stable RMSD values over time.

Commands:

```
./Run.prod1.cuda  
  
./Run.prod2.cuda  
  
./Run.prod3.cuda  
  
./Run.prod4.cuda  
  
./Run.prod5.cuda
```

Note: Production stages are being run sequentially, it is vital that the previous stages finish before running the next one.

4) Analysis:

RMSD & RMSF:

Both RMSD and RMSF can be analyzed using program *cptraj*. Script *Run.rms2* calculates the RMSD values throughout the whole loaded trajectory. It is important to monitor RMSD after each production state and it's nice to analyze the values throughout the whole simulation.

RMSF values can be calculated by running the *Run.tfactors2* script. RMSF can give us an indication on the fluctuation of the biomolecule.

The number of residues in both of the scripts needs to be adjusted to the analyzed complex. The values can be plotted for example with *xmgrace* or in some other way (eg. with a python script or in Excel).

Commands (RMSD):

```
./Run.rms2  
  
xmgrace [rmsd_file]
```

Commands (RMSF):

```
./Run.tfactors2  
  
xmgrace [rmsf_file]
```

Hydrogen bond distance analysis:

Since we're analyzing a complex of a receptor and ligands, it's also important to keep the interactions in mind. In particular, hydrogen bonds contribute to specific binding and anchoring the ligand in the binding site. Therefore, we monitored the distance between the atoms that were forming hydrogen bonds (according to PoseView and/or LigPlot). The atoms forming the hydrogen bonds need to be adjusted in the script.

Commands:

```
./Run.hbond_dist
```

```
xmgrace [distance_file]
```

References

1. Du, H. X. *et al.* Engineering *Yarrowia lipolytica* for campesterol overproduction. *PLoS One* **11**, (2016).
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3. Jumper, J. *et al.* Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, (2021).
4. Mirdita, M. *et al.* ColabFold: making protein folding accessible to all. *Nat Methods* **19**, (2022).
5. Trott, O. & Olson, A. J. AutoDock Vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *J Comput Chem* (2009) doi:10.1002/jcc.21334.
6. Pettersen, E. F. *et al.* UCSF Chimera - A visualization system for exploratory research and analysis. *J Comput Chem* **25**, (2004).
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8. Stierand, K., Maaß, P. C. & Rarey, M. Molecular complexes at a glance: Automated generation of two-dimensional complex diagrams. in *Bioinformatics* vol. 22 (2006).